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Artificial intelligence and financial fraud☆

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ABSTRACT

Corporate financial fraud significantly undermines stakeholder interests, weakens market resource allocation, and negatively impacts the healthy development of capital markets. This research leverages data from Chinese A-share listed companies (2013−2023), integrating machine learning techniques and text analysis to construct a firm-level artificial intelligence (AI) metric. Results indicate that advancements in AI technology significantly reduce both the likelihood and severity of financial fraud. This mitigating effect is especially pronounced in firms with separated chairman and CEO roles, those audited by non-Big Four firms, and larger enterprises. Mechanism analysis reveals that AI mitigates financial fraud mainly through enhanced information transparency and strengthened internal control systems. This study provides novel insights for policy-making and corporate practice.

1. Introduction

Financial fraud refers to a range deceptive practices undertaken by individuals or organizations to misappropriate assets or deliberately misrepresent financial information, thereby violating industry regulations and accounting standards. Such actions include, but are not limited to, asset misappropriation, financial statement manipulation, and false information disclosure (Yin et al., 2025). In recent years, despite continuous improvements in government regulation, legal frameworks, and external supervisory mechanisms, financial fraud incidents in listed companies have continued to occur frequently (Ajayi-Nifise et al., 2024; Ye et al., 2022). The provision of misleading financial information not only misguides stakeholders' decisions regarding economic activities in the market but also severely undermines the efficient allocation of resources, further disrupting investors' participation in capital market activities (Fich and Shivdasani, 2007; Hurwitz, 2019; Yin et al., 2025).

Artificial Intelligence (AI), a key driver of the current technological revolution and industrial transformation, has been widely integrated into corporate settings, facilitating both managerial efficiency and innovation (Fan et al., 2025). Its application in the financial domain has drawn significant attention (Cao et al., 2024). AI applies big data analytics, pattern recognition, and machine learning to improve the speed and accuracy of financial fraud detection, equipping stakeholders with advanced tools for prevention (Ali et al., 2022; Zhang et al., 2025). However, the relationship between AI implementation and corporate financial fraud, particularly the mechanisms through which AI influences fraudulent behavior, remains underexplored.

Many existing studies have explored the origins of corporate financial fraud from multiple perspectives, often grounding their

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0927-538X/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

analysis in the fraud triangle theory. This theory posits that financial fraud is typically driven by three key elements: pressure, opportunity, and rationalization. First, financial or performance pressures often drive fraudulent behavior (Biduri and Tjahjadi, 2024). Second, weaknesses in external supervision and internal control systems create opportunities for misconduct. Third, managers may justify unethical behavior by constructing rationalizations that ease internal conflict (De Klerk, 2017a). The fraud triangle theory offers a comprehensive framework for understanding how both internal and external factors jointly contribute to fraudulent behavior. Empirical studies have examined the influence of various factors, including executive characteristics (Jiang and Kim, 2015), macroeconomic conditions (Sun et al., 2023), and social oversight (Yang, 2022).

As AI evolves rapidly, its transformative tole in reshaping corporate management models and enhancing operational efficiency has become evident (Haefner et al., 2021). AI helps companies cope with market volatility by enabling precise demand forecasting, optimizing supply chains, and improving inventory management. In financial fraud detection and prediction, AI has been widely adopted from a preemptive regulatory standpoint to identify and forecast corporate misconduct (Bao et al., 2020; Al-Hashedi and Magalingam, 2021). Its use in auditing has also significantly enhanced efficiency and accuracy, thereby reducing the likelihood of fraudulent activities. Given that information opacity is a key factor in financial fraud (Broadstock and Chen, 2021), the integration of AI facilitates real time, automated data processing and enhances both internal communication and external disclosures (Zong and Guan, 2024).

Building upon existing literature, this study contributes to the field in three crucial dimensions.

Firstly, from the perspective of research focus: While prior studies have primarily examined the role of AI in financial fraud from an external viewpoint, emphasizing its function in predicting, identifying, and regulating fraud (Bao et al., 2020; Al-Hashedi and Magalingam, 2021). This study shifts the focus toward the internal application of AI within firms in addressing financial fraud. Although technologies such as supply chain digitalization (Jiang et al., 2024) and digital finance (Sun et al., 2023) has been well-explored, the potential of AI in areas like data analytics (Zdravković et al., 2022), anomaly detection (Hilal et al., 2022), and decision support (Bai and Li, 2024) has not been fully integrated into the discourse on fraud prevention. These technologies can be effectively deployed to prevent financial fraud at the enterprise level. The findings offer new insights for enterprises in formulating more effective financial governance strategies.

Secondly, concerning the mechanisms of influence: Whereas much of the existing literature relies on classical static models, such as the Fraud Triangle Theory (Abdullahi and Mansor, 2018) and the Five-Factor Fraud Theory, to explain fraud avoidance, this paper investigates how AI influences corporate financial fraud by reshaping the flow of internal and external corporate information under lens of information asymmetry, and introduces a dual-path mechanism: the information transparency mechanism and internal control mechanism. These mechanisms not only validate AI's comprehensive impact on the fraud triangle (i.e., pressure, opportunity, and rationalization), but also offer a novel analytical perspective for technology-enabled risk governance theory.

Thirdly, in terms of research methodology: This paper critically reviews prior approaches to constructing AI-related metrics, such as those based on robot penetration patent data (Du et al., 2024; Ye et al., 2023; Yang, 2022), and survey-based assessments (Cheng and Wu, 2023), highlighting their limitations. It then adopts a machine learning approach by applying Word2Vec technology and the Skip-gram model to analyze annual reports of listed companies, generating firm-level AI indicators through text mining (Mikolov et al., 2013). This novel and scalable approach enhances the precision and objectivity of AI measurement at micro level. The methodological framework employed herein may also be extended to other firm-level areas, such as stock price crashes and bond defaults.

2. Literature review

2.1. Causes of financial fraud

The fraud triangle theory posits that financial fraud is typically stems from the interplay of three primary factors: pressure, opportunity, and rationalization. Fraudulent behavior is likely to emerge only when all three conditions coexist (De Klerk, 2017).

Pressure constitutes the internal driving force behind fraudulent behavior (Shi et al., 2017). It often arises from both economic and psychological stressors. Specifically, individuals or firms may experience financial distress, high performance expectations, resource constraints, or aspirations for elevated social status and power (Biduri and Tjahjadi, 2024). In such contexts, fraudulent behavior may serve as a means to temporarily alleviate these pressures.

Opportunity refers to external conditions that facilitate fraudulent behavior. These include weak internal controls (Donelson et al., 2017), lenient penalties (Wang et al., 2022), significant information asymmetry (Broadstock and Chen, 2021), inadequate regulatory oversight, low audit quality (Biduri and Tjahjadi, 2024), and flawed audit systems. Such conditions enable perpetrators to act with reduced rish of detection and punishment.

Rationalization represents the psychological process by which individuals justify unethical or illegal actions. Faced with pressure and the opportunity, fraudsters often construct self-justifying narratives (e.g., "temporary fraud to save the company") to ease their internal moral dissonance and legitimize misconduct (Bandura, 1986). This rationalization process facilitates the decision to commit fraud and completes the motivational framework described in the fraud triangle.

2.2. Determinants of financial fraud

At the macro level, Hou et al. (2021) employed a bivariate probit model to demonstrate that firms tend to be more prone to commit financial fraud when economic policy uncertainty is elevated. Zhao et al. (2024), using a difference-in- differences model on Chinese listed companies, found that stricter environmental regulations effectively reduce both the motivation and frequency of financial

fraud. Al-Shattarat and Al-Shattarat (2021) examined survey data from Jordanian commercial banks listed on the Amman Stock Exchange and reached the conclusion that a well-established regulatory environment enables stakeholders to more effectively detect ant prevent fraudulent behavior. Zhang (2018), drawing on enforcement data from the China Securities Regulatory Commission (CSRC) between 2004 and 2014, found that stronger anti-corruption measures and higher public governance standards exerts a significant inhibitory effect on corporate financial fraud. Sun et al. (2023) further showed, using panel data, that broader digital finance coverage and deeper utilization significantly curb the incidence of financial fraud among Chinese listed firms.

At the micro level, many scholars have explored the drivers of financial fraud within listed companies from the perspectives of shareholders, managers, and employees. Zingales (1994) and Bae et al. (2002) argued that dominant shareholders may pursue private benefits by diverting company assets, thereby harming minority shareholders and triggering internal fraud. Johnson et al. (2000) further explained that such expropriation often manifests through direct misappropriation, unwarranted guarantees, and related-party transactions, all of which exacerbate legal violations. Beasley (1996)examined board composition and found that a higher proportion of independent directors correlates with a reduced likelihood of financial fraud. Kim et al. (2022) revealed that managerial ownership substantially influences unethical behavior, especially in financially distressed companies where managers may engage in misconduct to boost short-term performance. Independent directors also play an important role in corporate governance: Bravo and Reguera-Alvarado (2018) suggested that their effectiveness increases with board tenure, while Dimmock et al. (2018) found that the personal credit among employees in financial advisory firms may elevate fraud risks. Finally, Wu et al. (2023) pointed out that employee stock ownership plans can reduce managerial opportunism, enhance both internal and external oversight, thereby mitigating the occurrence of fraud.

2.3. The impact of artificial intelligence on firms

As a driving force in the current wave of technological and industrial transformation, AI exerts profound impacts across various dimensions of corporate operations. Existing research mainly focuses on the beneficial effects of AI across total factor productivity, corporate innovation, and industrial restructuring.

In terms of total factor productivity (TFP), Damioli et al. (2021), using patent data from 5257 global companies, found that AI patents significantly enhance firms-level productivity, particularly among small and medium-sized enterprises (SMEs) and within the service sector. Brynjolfsson et al. (2019) emphasize the TFP gains from AI through improved task automation and data-driven process optimization, offering theoretical basis for productivity gains in technology-intensive environments. Yang and Liu (2024), through text analysis of annual reports and patent disclosures from Chinese listed firms, concluded that AI substantially boosts TPF in the manufacturing sector, especially under conditions of high industry concentration. Chang et al. (2023), based on provincial panel data, identified a significant positive marginal effect of AI on green TFP. Furthermore, Yang and Liu (2024) observed, using city-level panel data, that AI's positive impact on green TFP depends heavily on effectiveness of regulatory environments. Chang et al. (2023) also argued that AI promotes green productivity by reducing energy consumption, particularly in labour-intensive and technology-intensive industries.

In terms of corporate innovation, Han and Mao (2024) employed patent and robotics data to empirically demonstrate that AI accelerates firms' green innovation by upgrading workforce skills, increasing R&D investment, reducing labour costs, and promoting information sharing and resource coordination. Tian et al. (2023), based on panel data analysis, further revealed that AI enhances firms' green technology innovation by strengthening their knowledge integration capacity. **Cockburn et al. (2018)** also suggest that AI fundamentally reshapes the innovation landscape by enabling new forms of invention and expanding the boundaries of technological capabilities. Qu and Li (2022) and Bai and Li (2024) found that AI, empowered by big data analytics, provides firms with data-informed decision support, thereby enhancing innovation management. Su et al. (2024) analysed AI's impact across different corporate development stages and found that AI significantly drives green innovation in the start-up, growth, and maturity phases by optimization resource allocation and reducing operational coasts.

In terms of industrial structural transformation, Bresnahan and Trajtenberg (1995) suggested that AI technology propels traditional industries toward higher level of technological intensity, thereby fostering industrial restructuring and optimization. Cao et al. (2024) noted that as AI becomes increasingly pervasive in areas such as smart home systems, autonomous vehicles, and intelligent healthcare, traditional economic models are undergoing a rapid transformation toward digitalization and intelligence. Frey and Osborne (2017) argued that AI would reshape labour market structures by displacing certain traditional roles while simultaneously creating new occupational categories. Arntz et al. (2016) further emphasized that AI adoption varies across sectors and job types, leading to a differentiated impact on employment, with middle-skill routine jobs most exposed to automation risk across OECD countries. Fan et al. (2025), through empirical analysis of provincial panel data, found that AI accelerates regional economic growth by facilitating structural adjustment. Wei et al. (2021) demonstrated that AI significantly contributes to optimization industrial structures in China, particularly in high-tech manufacturing and producer service sectors.

2.4. Literature review summary

This research revies the existing literature on financial fraud and the influences of AI on enterprises. Firstly, the fraud triangle theory provides a robust theoretical framework that explains how both internal and external conditions lead to fraudulent behavior, focusing on the dimensions of pressure, opportunity, and rationalization. Both empirical evidence and case studies support the theory's applicability, examining how fraud emerges under specific organizational, regulatory and behavioral conditions. Secondly, with respect to the impact of AI, the literature generally emphasizes its positive role in enhancing productivity, advancing green

development, fostering corporate innovation, and driving industrial restructuring. From the standpoint of fraud prevention, prior studies largely center on AI's role in passive monitoring, particularly through anomaly detection techniques, bigdata analytics and machine learning-based classification methods (Bao et al., 2020; Al-Hashedi and Magalingam, 2021; Lin, 2024). However, much of the literature focuses on external regulatory mechanisms and their fraud-suppressing effects, while discussion on how AI interacts with internal financial misconduct remain limited. This study aims to address this gap by exploring how the repaid advancement of AI technology can proactively shape corporate financial behavior by improving transparency, optimizing decision-making processes, and reinforcing internal control systems.

3. Theoretical analysis and research hypotheses

3.1. Direct impact of corporate artificial intelligence on financial fraud

This study is grounded in fraud triangle theory proposed by Abdullahi and Mansor (2018), which remains a foundational model in financial misconduct research. The theory posits that financial fraud stems from the interplay of three elements: pressure, opportunity, and rationalization. While the theory has been widely supported by empirical studies, few studies have examined how technological advancement, particularly AI, affects theses three drivers of fraud.

This study argues that the application of AI reshapes the traditional fraud triangle framework by redefining the structure and relative influence of pressure, opportunity, and rationalization. By embedding intelligent systems into internal corporate processes, AI not only changes how decisions are made and monitored but also transforms the underlying behavioral incentives and ethical constraints that lead to fraud. **Wamba-Taguimdje et al. (2020)** emphasize that AI affects not just detection but also organizational structure and internal awareness, thus playing a transformative role in shaping fraud-related decision environments.

First, regarding the pressure factor, significant fluctuations and uncertainties in the external environment often cause enterprises to deviate from their expected operational goals and strategic plans, leading to substantial financial and performance pressure. Such pressure may drive some members of management to engage in fraudulent activities in pursuit of short-term benefits. In this context, enterprises can leverage AI-related technologies to enhance their adaptive response to market changes (ARMC), thereby creating organizational value and addressing various pressures arising from market fluctuations. For example, through machine learning and big data analytics, enterprises can more accurately predict strategies, including operational scheduling and dynamic resource allocation, and apply predictive maintenance in large-scale manufacturing systems. These measures help optimize resource allocation and production efficiency, thereby reducing the pressure caused by performance fluctuations (Sun et al., 2024a). This adaptive response capability not only enables enterprises to proactively adjust strategies and relocate resources but also effectively alleviates internal pressures caused by environmental turbulence and performance instability, thus fundamentally reducing the likelihood of fraud driven by pressure. **Chen et al. (2021)** further argue that AI contributes to organizational resilience by enhancing real-time operational responsiveness and supporting rapid strategic reconfiguration under uncertainty.

Secondly, from the perspective of opportunity, fraudulent activities frequently arise from weak internal controls and inadequate supervision. The widespread application of AI serves as a robust informational safeguard for enterprises, enhancing internal control systems through automated and intelligent auditing processes. By integrating intelligent auditing systems and real-time data monitoring technologies, businesses can achieve end-to-end tracking of financial transactions, enabling the prompt identification of anomalies and potential risks (Hilal et al., 2022). **Moffitt and Vasarhelyi (2013)** highlight that AI-enabled continuous auditing significantly improves the timeliness and reliability of internal control procedures, thereby minimizing the time lag between fraudulent activity and its detection.

Machine learning algorithms demonstrate significant advantages in analyzing historical data and detecting anomalies, enabling rapid detection and flagging or irregular financial activities. This not only substantially decreases the probability of fraudulent behavior but also enhances transparency and regulatory compliance in internal operations, strengthening the effectiveness of internal audits and reinforcing corporate governance.

Finally, from the perspective of rationalization, when management engages in fraudulent activities, they often construct self-justifying narratives (e.g., "temporary fraud to save the company") to justify their actions. AI directly disrupts these psychological mechanisms by enhancing decision-making transparency and supervisory deterrence. On the one hand, intelligent auditing systems and risk management tools ensure that all financial data flows and decision-making processes are fully traceable (Hilal et al., 2022). Every operation leaves a digital footprint, making it difficult to conceal misconduct and thereby preventing management from constructing rationalization excuses. **Tenbrunsel and Messick (2004)** emphasize that AI-generated transparency can interrupt "ethical fading", a cognitive distortion that enables actors to disengage from moral accountability—thus challenging the rationalization foundation of fraudulent decision-making. On the other hand, the persistent deterrence effect created by real-time monitoring systems forces management to reassess the cost-benefit tradeoff of fraud (Immadisetty, 2025). This technological constraint fundamentally alters management's risk preferences, weaking the psychological mechanism of moral disengagement (Bandura, 1986) that would otherwise facilitate unethical behavior. A highly transparent and traceable governance mechanism compels management to confront the company's actual financial status, thereby reducing the space for self-rationalization and ultimately diminishing the risk of fraud driven by rationalization.

Therefore, based on theoretical analysis, this study proposes the following research hypotheses:

H1: Artificial intelligence can significantly inhibit corporate financial fraud.

3.2. Theoretical mechanisms of artificial intelligence on corporate financial fraud

3.2.1. Information transparency mechanism

With the rapid development of AI, an increasing number of AI technologies have been applied in companies, enabling firms to efficiently access both internal and external information, promote operational transparency, reduce information asymmetry, and improve overall information disclosure.

In terms of the overall volume of information disclosure, AI has revolutionized the traditional model of data processing model by converting large volumes of unstructured or ineffective data into visualized and standardized formats, thereby activating internal data dynamics within the organization (Wang et al., 2022). At the same time, AI provides a new platform for the exchange of internal and external information, overcoming the constraints of resource and information flows and significantly increasing information dissemination. Regarding the quality of information disclosure, AI has driven a transformation in corporate management models by establishing real-time monitoring mechanisms based on digitized business scenario data. It automates the processing and analysis of complex financial and operational data, reducing managerial friction, improving corporate governance, and enhancing the accuracy and transparency of information. In terms of readability, due to the positive "exposure effect," companies, especially those seeking financial support from the capital market, tend to convey newly acquired technological insights with investors and present more intelligible repots, thereby further enhancing the comprehensibility of disclosed content (Shiyyab et al., 2023; Wang et al., 2022).

Increased information transparency reduces opportunity for financial fraud. Management may conceal negative information, especially financial data, to protect their personal interests, such as maintaining job security, inflating the value of stock options, or pursuing political advancement. Information asymmetry impairs stakeholder's ability to promptly detect such concealment, thereby creating conditions conducive to fraudulent behavior (Ndofor et al., 2015). Within firms, the implementation of AI ensures that information is captured, recorded, and processed in real time throughout its entire lifecycle, from generation to disclosure, enabling early detection of anomalies. This reduces on manual financial processing and limits opportunities for data manipulation and fraud. AI also assists in integrating information, effectively addressing legacy issues of information overload and ensuring that discussion-makers across all levels of the organization complete the accurate information. This enhances managerial decision-making quality and reduces the risk of fraud stemming from poor judgment. Externally, AI improves the transparency of business operations and strengthens the quality of audit evidence, thereby elevating audit quality. **Tang et al. (2021)** further note that AI reduces disclosure distortions by enhancing automated data cleansing and reporting accuracy, especially in complex, data-intensive firms. As more financial and op, rational information is shared externally, stakeholders are better equipped to monitor corporate behavior, ultimately reducing the incidence of fraudulent activities.

H2: Corporate artificial intelligence can suppress financial fraud by enhancing information transparency.

3.2.2. Internal control mechanism

AI can enhance corporate internal control systems in multiple ways, notably by improving the efficiency and quality of internal audits and managerial decision-making. In the domain of internal auditing, AI can automate repetitive tasks such as data collection and preliminary analysis, allowing accounting and financial personnel to focus on higher-value tasks, thus elevating overall audit efficiency (Fedyk et al., 2022). Furthermore, AI can automatically summarize and deliver relevant information to decision-makers, tailoring disclosures to stakeholder behavior characteristics. Integrating AI into internal control systems strengthens mutual oversight and balances internal power structures, reducing human subjectivity in internal control execution.

Through deep learning and machine learning algorithms, AI technology continuous real-time tracking and monitoring of enterprise data, allowing for the prompt detection of irregularities and supporting more informed and evidence-based decision-making (Johnson et al., 2000). In the internal control goal-setting process, AI's powerful data processing capabilities enable the acquisition of broader external intelligence, helping firms establish control objectives that respond dynamically to both internal and external environments. This allows companies to avoid passive responses to market fluctuations (Zdravković et al., 2022).

Internal control is a key mechanism for preventing financial fraud. Weak internal control systems or poor implementation can create significant opportunities for fraud (Donelson et al., 2017). Robust internal control systems provide clear division of responsibilities, enforce authorization procedures, and incorporate strong supervisory mechanisms. These features allow for the timely detection and correction of violations, thereby reducing the likelihood of fraudulent activity. Conversely, poorly designed or ineffective internal controls may allow managers to override safeguards, facilitating fraud. In numerous real-world cases, the absence of comprehensive internal evaluation mechanisms delays the detection of fraudulent transactions. The convergence of ownership and management authority weakens board oversight, while a lack of independence among internal audit department undermines their supervisory role. **Gepp et al. (2018)** further demonstrate that AI-enhanced auditing systems outperform traditional approaches in early-stage fraud detection and in supporting audit independence. These factors, when unresolved, collectively contribute to internal control failure, and, subsequently, increased fraud risk (Kabuye et al., 2017).

Therefore, we propose the following hypothesis:

H3: Corporate artificial intelligence can suppress financial fraud by enhancing internal control.

4. Research design

4.1. Data sources

This research selects Chinese A-share listed companies from 2013 to 2023 as its research sample. The year 2013 is chosen as the

starting point because the number of financial fraud cases in China has experienced a significant increase since 2014 (Ye et al., 2022). The sample data are obtained from the Wind Database and the CSMAR Database. To ensure the validity of the sample, the following data-cleaning procedures were applied (Kong et al., 2022): (1) Firms with ST, *ST, and PT designations were excluded; (2) Companies in the financial industry were removed; (3) Firms with substantial missing values were eliminated; (4) Companies operating in sectors such as software and information technology services were excluded. These industries are highly dependent on AI-related technologies and tend to disclose relevant information directly in their annual reports, making it difficult to accurately evaluate the incremental impact of AI adoption. After applying theses filters, the final sample consists of 26,182 firm-year observations. To mitigate the influence of outliers, all continuous variables were winsorized at the 1 % level and 99th percentiles. Liu et al. (2020) adopt a similar sampling and winsorization method for A-share firm analysis, ensuring comparability and methodological consistency across Chinese financial fraud studies.

4.2. Variable definitions

4.2.1. Dependent variables

Based on existing studies (Sun et al., 2023), this paper constructs relevant indicators using corporate violation data from the CSMAR database. Listed companies classified under the violation types of "inflated assets," "fictitious profits," "delayed disclosure," "material omissions," and "false statements" are designated as part of the financial fraud sample. A dummy variable (*Fdum*) is constructed to represent the occurrence of financial fraud: If a company is found to have committed financial fraud and is concurrently recognized as violating regulations by one or more regulatory bodies(Yin et al., 2025), the value of *Fdum* is set to "1"; otherwise, it is set to "0."

Additionally, this paper incorporates an ordinal variable for fraud severity (Fdeg), following the approach of Sun et al., 2023. If a listed firm incurs no penalty, Fdeg is given a value of "0." In only executives are penalized and the company is not, or if both are penalized but the company's penalty falls under the "other" category, then the value of Fdeg is set to "1." If a firm receives faces a penalty of "criticism" or "censure," the value is "2." If the firm is penalized with "warnings, fines, or confiscation of illegal profits," Fdeg is assigned a value of "3." In cases where multiple penalty types occur within the same year, the highest level of fraud severity is adopted.

4.2.2. Key independent variables

Previous studies on AI-related indicators at the firm level have typically focused on three methodological approaches. First, robot-related metrics have been widely used, including robot penetration (Du et al., 2024), the number of robot installations (Ye et al., 2023), and the per capita value of machinery and equipment. However, this paper does not adopt these approaches, as modern AI development today extends far beyond robotics. Technologies such as GPT models, smart recognition, and "Internet+" platforms are also integral to AI application in enterprises. Relying solely on robot-based indicators risks overlooking firms that utilize non-robot AI technologies. Second, patent-based measurements have been employed (Yang, 2022), but this method is not adopted here due to the strategic or confidential nature of some innovations, which may defer firms for applying for patents. Third, survey-based methods have been used (Cheng and Wu, 2023), but given the limited number of surveys and the potential biases introduced by survey quality and respondent mood, this approach may compromise data reliability.

To construct a more precise and objective enterprise - level AI indicators, this research selects the annual reports of Chinese A-share listed companies as textual data sources. Applying machine learning and text analysis techniques, specifically the Skip-gram model within Word2Vec technology, this method captures sematic and contextual information in company reports from 2013 to 2023. **Li and Wang (2021)** demonstrate the effectiveness of Word2Vec in constructing textual indicators for firm-level digitalization, validating its adaptability in analyzing Chinese financial documents. Compared to traditional machine-learning models (e.g., Support Vector Machines or Naive Bayes), the Skip-gram model automatically learns from large textual corpora, reducing the complexity and subjectivity with manual feature extraction. Word2Vec transforms each word into a high-dimensional vector reflecting its semantic and contextual characteristics. By calculating cosine similarity between vectors, the Skip-gram model identifies sematic relationships words, enabling aI-related feature learning without extensive manual preprocessing. Its efficiency and scalability make it well-suited for processing large volumes of text, such as the annual reports from 2013 to 2023. Therefore, the Skip-gram model enables more efficient and accurate identification of AI-related keywords. The specific construction process of the AI indicator is as follows (see Fig. 1):

(1) Data Collection and Processing: Annual reports of all listed Chinese companies from 2013 to 2023 are retrieved from the Cninfo

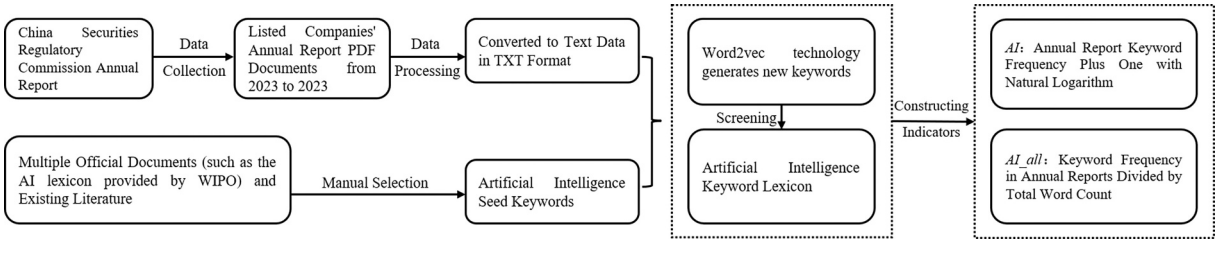


Fig. 1. Process of Constructing Enterprise Artificial Intelligence Indicators.

platform and converted into UTF-8 encoded text files, totaling 40,916 documents. (2) Selection of Seed Words: Based on AI lexicon provided by the World Intellectual Property Organization (WIPO) and existing literature (Li and Wang, 2022), 25 AI-related seed words are manually selected, including terms such as "robot," "industrial internet," "AI," and "digitalization." (3) Machine Learning with Word2Vec Technology: The Skip-gram model from Word2Vec (Mikolov et al., 2013) is trained on the full corpus. Cosine similarity is used to identify the 15 words most semantically related terms for each seed word. To address segmentation errors common in Chinese texts (e.g., splitting "AI" into "artificial" and "intelligence"), Jieba segmentation tool is employed with a custom dictionary and stop-word filtering. (4) Manual Cleaning: Duplicate, irrelevant, or low-frequency terms are manually removed, yielding a final keyword list of 58 AI-related terms. (5) Text Segmentation and Keyword Frequency Calculation: Using Jieba, all custom dictionary, and stop words are removed. The frequency of AI-related terms keywords and the total word count of each annual report are recorded. (6) Construction of AI Indicators: AI: The natural logarithm of (AI keywork frequency + 1) is used as the main indicator. AI_all: an alternative indicator divides AI keyword frequency by total word count the report. This approach enables a robust, scalable, and objective measurement of firm-level AI intensity based actual textual disclosures.

4.2.3. Control variables

This research includes control variables according to the existing literature and classifies them into three levels: firm-level, entrepreneur-level, and external firm-level variables (Faff et al., 2019). (1) Liquidity is measured as the ratio of net cash inflow to current liabilities; (2) Size, Logarithmic value of total assets of the enterprise at the end of the year; (3) State ownership (*Soe*); (4) The book-to-market ratio (*Btm*) is defined as the ratio of total assets to market value; (5) Duality (*Dual*) equals 1 if the chairman and CEO are the same individual and 0 otherwise; (6) Board size (*Bdsize*); (7) Proportion of independent directors (*Indep*), measured as the ratio of independent directors to the total number of board members; (8) Ownership concentration (*First*), defined as the proportion of shares held by the largest shareholder; (9) Big4 equals 1 if the firm is audited by one of the Big Four auditing firms and 0 otherwise. The definitions of these variables and their descriptive statistics are provided in Table 1.

4.2.4. Mechanism variables

The information transparency index (*Opacity*) in this research is constructed based on the "Transparency of Listed Companies" rating documents published by the Shenzhen Stock Exchange (*SZSE*) and the Shanghai Stock Exchange (*SSE*). The internal control index (*In_Control*) is derived from each listed company's internal control evaluation report. Specifically, the "Transparency of Listed Companies" ratings are converted into the following numerical scale to contract the information transparency index (*Opacity*):from the SZSE and SSE are assigned values of 4 for "Excellent," 3 for "Good," 2 for "Pass," and 1 for "Fail". Similarly, the internal control evaluation reports disclosed by companies are used to generate the internal control index (*In_Control*) based on the following criteria: a value of 3 is assigned for "Effective" internal control with "no significant defects," 2 for "Effective" internal control with "defects," and 1 for "Ineffective" internal control, These scoring rules ensure the construction of consistent and quantifiable mechanism variables to examine how AI influences financial fraud through improvements in information transparency and internal control quality.

4.3. Model construction

To examine the impact of the corporate AI on financial fraud behavior, this study constructs the following model for the basic regression analysis:

$$\text{Fraud}_{it} = \beta_0 + \beta_1 AI_{it} + \beta_2 Control_{it} + yeardum + inddum + \varepsilon_{it} \quad (1)$$

To verify that corporate AI affects financial fraud through information transparency, this paper constructs models (2) and (3):

To test Hypothesis 2, namely that artificial intelligence influences corporate financial fraud through information transparency, this paper constructs the following models (Sun et al., 2024b):

$$Opacity_{it} = \beta_0 + \beta_1 AI_{it} + \beta_2 Control_{it} + yeardum + inddum + \varepsilon_{it} \quad (2)$$

Table 1
Descriptive.

variables	N	mean	sd	min	max
Fdeg	26,182	0.231	0.591	0	3
Fdum	26,182	0.165	0.372	0	1
AI	26,182	2.509	1.259	0	5.855
AI_all	26,182	0.002	0.003	0	0.023
Big4	26,172	0.065	0.246	0	1
Liquidity	26,182	0.246	0.396	−0.615	2.052
Dual	25,443	0.288	0.453	0	1
First	26,172	34.795	14.772	8.48	74.02
Soe	26,172	0.351	0.477	0	1
Size	26,182	22.3	1.332	19.871	26.398
Bdsize	26,171	2.121	0.193	1.609	2.639
Indep	26,171	37.564	5.299	33.33	57.14
Btm	26,182	0.64	0.251	0.112	1.192

$$Fraud_{it} = \beta_0 + \beta_1 Opacity_{it} + \beta_2 Control_{it} + yeardum + inddum + \varepsilon_{it} \quad (3)$$

To examine whether corporate AI affects financial fraud through internal control, this paper constructs Models (4) and (5):

To test Hypothesis 3, which posits that artificial intelligence affects financial fraud by influencing corporate internal control, the following models are developed:

$$In_Control_{it} = \beta_0 + \beta_1 AI_{it} + \beta_2 Control_{it} + yeardum + inddum + \varepsilon_{it} \quad (4)$$

$$Fraud_{it} = \beta_0 + \beta_1 In_Control_{it} + \beta_2 Control_{it} + yeardum + inddum + \varepsilon_{it} \quad (5)$$

The dependent variable, Fraud, represents corporate financial fraud, and includes both the binary fraud indicator (*Fdum*) and the degree of fraud (*Fdeg*). The subscripts *i*, and *t* represent the firm and the year, respectively. When the dependent variable is *Fdum,* the logit model is employed for estimation; when the dependent variable is *Fdeg,* the OLS regression model with multidimensional fixed effects is employed. The core explanatory variable, AI, represents the level of corporate AI adoption. The mechanism variables, Opacity and In Control, correspond to the information transparency indicator and internal control indicator, respectively, and are both constructed using text analysis techniques. *Control* variables include factors at the firm level, entrepreneur level, and external level, as defined in Section 3.2.3.

5. Empirical analysis

5.1. Baseline regression analysis

Table 2 presents the baseline regression findings exploring the relationship between corporate AI adoption and financial fraud. Columns (1) and (2) report the estimated impact of AI on the degree of financial fraud. Meanwhile, columns (3) and (4) display the estimated impact of AI on the occurrence of financial fraud. In column (1), the coefficient of AI is −0.013, and is statistically significant at the 1 % level. In column (2), after controlling additional control variables, the coefficient of AI remains significantly negative. When the dependent variable is changed to corporate fraudulent behavior (Fdum), the absolute value of the AI coefficient increase to −0.037 and − 0.063 in columns (3) and (4), respectively, and both coefficients are also significant at the 1 % level. This indicates that a higher degree of corporate AI adoption can significantly reduce the likelihood and severity of financial fraud, thereby providing empirical support for hypothesis H1. This effect attributed to AI's ability to help firms adapt to changing market changes through advanced data analysis and predictive modeling. These technologies enhance financial planning capabilities, alleviate financial pressure, and improve operational efficiency. Furthermore, through intelligent real-time monitoring and data analysis, AI minimizes human intervention, identifies potential risk points and abnormal behaviors, and enables companies to detect and response to emerging fraud risks a timely

Table 2
Baseline Regression Results.

variables	(1)	(2)	(3)	(4)
	Fdeg	*Fdeg*	*Fdum*	*Fdum*
AI	−0.013***	−0.014***	−0.037***	−0.063***
	(−3.37)	(−3.61)	(−2.83)	(−4.60)
Big4		−0.079***		−0.649***
		(−6.26)		(−6.61)
Liquidity		−0.160***		−0.802***
		(−16.78)		(−15.21)
Dual		−0.021**		−0.125***
		(−2.42)		(−3.12)
First		−0.004***		−0.019***
		(−14.79)		(−14.54)
Soe		−0.067***		−0.310***
		(−6.96)		(−7.19)
Size		0.011***		0.057***
		(3.02)		(3.17)
Bdsize		−0.119***		−0.470***
		(−4.49)		(−4.00)
Indep		−0.002***		−0.006
		(−2.64)		(−1.49)
Btm		−0.074***		−0.381***
		(−3.80)		(−4.56)
Constant	0.264***	0.615***	−1.526***	−0.304
	(25.11)	(6.55)	(−41.61)	(−0.66)
Industry	yes	yes	yes	yes
Year	yes	yes	yes	yes
N	26,182	25,443	26,182	25,443

Notes: ***, **, * denote statistical significance at the 1 %, 5 %, and 10 % levels, respectively. Values in parentheses are t-value calculated from robust standard errors. Unless otherwise specified, the same notation applies to all subsequent tables.

manner, ultimately lowering the probability of fraudulent conduct.

5.2. Robustness test

(1) Based on the company's emphasis on AI, this study re-measured the level of corporate AI by calculating the ratio of AI-related keywords frequency to the total word count in each firm's annual report. The regression outcomes are presented in columns (1) and (2) of Table 3. For both the degree of corporate financial fraud (*Fdeg*) and the incidence of financial fraud (*Fdum*), the regression coefficients are negative and statistically significant at the 1 % and 5 % levels respectively. This result confirms that the conclusions of this research remain valid. (2) Lagging the explanatory variable helps mitigate potential endogeneity problem caused by reverse causality. Thus, this paper conducts a test using lagged AI variables. Thus, outcomes, presented in columns (3) and (4) of Table 3, are consistent with the baseline regression results, further validating the findings. (3) To assess robustness under a shortened sample period, this study considers policy implications. The14th Five-Year Plan, introduced in 2021, highlighted that new-generation information technologies represented by AI will serve as a core driver of national innovation. Since then, AI development has entered a new phase. To avoid potential bias arising from policy impacts, the sample is restricted to the pre-2021 period. The outcomes, presented in columns (4) and (5) of Table 3, remain consistent with the baseline regression, confirming the robustness of the findings.

5.3. Endogeneity test

To address potential endogeneity caused by sample selection and omitted variable bias, this study adopts the Propensity Score Matching method (PSM) for further verification. Considering that the AI indicator is a continuous variable, it is dichotomized based on the sample median. Enterprises with an AI word frequency greater than the median are categorized as high-AI firms (assigned a value of 1), while those below the median are defined as Low-AI firms (assigned a value of 0), serving as the control group. The Propensity Score Matching (PSM) method is applied using the control variables as covariates, and the sample is matched via 1:1 nearest-neighbor matching with replacement. After matching, the standardized differences between the treatment and control groups are reduced, with most falling below 5 %. The *t*-test results further confirm that there are no statistically significant differences in covariates post-matching. This implies that the PSM procedure effectively balances the sample. A re-regression using the matched samples is conducted, and the results, reported in Table 4, show that the AI variable remains significantly negative at the 1 % level. These results demonstrate that, after addressing self-selection concerns, the main findings of the study remain valid and robust.

5.4. The impact of artificial intelligence on the mechanism of corporate financial fraud

The aforementioned research indicates that a higher level of corporate adoption is associated with a lower likelihood of financial fraud. Based on the theoretical framework and hypotheses discussed earlier, this section investigates the mechanisms through which AI influences corporate financial fraud. The two key mechanisms explored are information transparency and internal control.

5.4.1. Information transparency mechanism

Corporate information transparency reflects the degree to which firms disclose crucial information, such as operational performance and financial conditions, to external stakeholders. High transparency entails frequent and accurate public disclosure of financial statements and operational updates to investors, regulators, and the general public. This increases the probability that fraudulent behavior will be detected. When firms are aware that their financial activities are under closer scrutiny, both the incentives and opportunities for management to engage in fraudulent activities are curtailed. Consequently, enterprises with high-level information transparency are notably less likely to engage in financial misconduct (Ndofor et al., 2015). This paper utilizes Models (2) and (3) to empirically text the transparency mechanism. The regression results are presented in Table 5.

The findings in column (1) indicate that the overall impact of AI on corporate information transparency is 0.057, and this result is statistically significant at the 1 % level. This suggests that AI significantly enhances corporate transparency. AI may help enterprises better demonstrate the authenticity and integrity of their financial status and business operations by leveraging advanced data processing, real-time information disclosure support, and intelligent report generation. The result in column (2) further verify this conclusion: even without control variables, the positive relationship between AI and information transparency remains significant. The results presented in columns (3) to (6) imply that, regardless of the addition of control variables, higher information transparency

Table 3
Robustness Test Results.

variables	(1)	(2)	(3)	(4)	(5)	(6)
	Fdeg	*Fdum*	*Fdeg*	*Fdum*	*Fdeg*	*Fdum*
AI			−0.011**	−0.063***	−0.020***	−0.033**
			(−2.43)	(−4.28)	(−4.14)	(−2.05)
AI_all	−0.379**	−20.721***				
	(−2.34)	(−3.51)				
Control , Industry,Year	yes	yes	yes	yes	yes	yes
N	25,424	25,424	21,302	21,302	18,333	18,333

Table 4
Endogeneity Test Results.

variables	(1)	(2)
	Fdeg	*Fdum*
PSM_AI	−0.013***	−0.006***
	(−2.59)	(−3.50)
Control	yes	yes
Industry	yes	yes
Year	yes	yes
N	25,434	25,434

Table 5
Test Results for the Information Transparency Mechanism.

variables	(1)	(2)	(3)	(4)	(5)	(6)
	Opacity	*Opacity*	*Fdeg*	*Fdeg*	*Fdum*	*Fdum*
AI	0.057***	0.082***				
	(13.47)	(18.83)				
Opacity			−0.339***	−0.338***	−0.195***	−0.199***
			(−34.68)	(−37.82)	(−40.50)	(−45.72)
Control	yes	no	yes	no	yes	no
Industry	yes	yes	yes	yes	yes	yes
Year	yes	yes	yes	yes	yes	yes
N	20,077	20,077	20,077	20,077	20,077	20,077

significantly reduce both the degree and occurrence of financial fraud, with all coefficients significant at the 1 % level. It further illustrates that improved information disclosure reduces the opportunities and incentive for managers to manipulate financial statements by increasing the visibility of of corporate activities. When transparency is high, stakeholders, investors, and regulators can detect irregularities more effectively and in a timelier manner, thereby mitigating the risk of financial fraud. There findings provide empirical support for Hypothesis H2.

5.4.2. Internal control mechanism

The principle of comprehensiveness in internal control mandates that enterprises implement control mechanisms throughout the entire process of decision-making, execution, and supervision. This helps to eliminate operational blind spots and systemic gaps, thereby mitigating the likelihood of fraudulent behavior. The principle of checks and balances ensures that power and responsibility are distributed across departments, so no single party retains excessive authority, thus effectively preventing misuse and fraudulent actions. A robust internal control framework provides a reliable operating environment for enterprises by ensuring the legality, safety, efficiency, and effectiveness of their management activities, thereby reducing the risk of fraudulent activities. Numerous studies have validated the crucial role of internal control in mitigating corporate financial fraud (Donelson et al., 2017; Tian and Sun, 2023). After removing enterprises that did not disclose internal control evaluation reports, this paper employs Models (4) and (5) to test internal control mechanism. The regression results are shown in Table 6.

The findings in column (1) show that the overall influence of AI on corporate internal control is 0.002, and the result is statistically significant at the 1 % level. This indicates that AI exerts a remarkably positive effect on strengthening internal control systems. As noted earlier, AI effectively boosts the efficiency and position of internal control through real-time data analysis, intelligent process monitoring, and anomaly detection. Appelbaum et al. (2017) further support that AI-enabled continuous auditing mechanisms can improve anomaly identification and overall governance robustness.

The results in column (2) further confirm this conclusion, AI's impact remains significant even in the absence of control variables.

Table 6
Test Results for the Internal Control Mechanism.

variables	(1)	(2)	(3)	(4)	(6)	(7)
	Opacity	*Opacity*	*Fdeg*	*Fdeg*	*Fdum*	*Fdum*
AI	0.002***	0.001**				
	(2.62)	(2.14)				
In_Control			−1.613***	−1.679***	−0.651***	−0.659***
			(−15.40)	(−16.36)	(−20.30)	(−22.00)
Control	yes	no	yes	no	yes	no
Industry	yes	yes	yes	yes	yes	yes
Year	yes	yes	yes	yes	yes	yes
N	15,852	15,852	15,852	15,852	15,852	15,852

Regardless of whether control variables are included, internal control significantly reduces both the degree and occurrence of financial fraud, with all coefficients significant at the 1 % level. This confirms that internal control mechanism is effective and reliable. The effectiveness of internal control can be observed through several mechanisms: effective risk assessment, well-defined responsibilities, and stringent process supervision all contribute to fraud mitigation. These findings empirically verify Hypothesis H3.

5.5. Heterogeneity analysis

The advancement of AI technology plays a vital role in enhancing corporate information transparency and strengthening internal control systems, thus contributing to the reduction of financial fraud within enterprises. However, due to substantial differences in corporate organizational structures, external regulatory environments, and their inherent characteristics of firms, the inhibitory effect of AI on financial fraud may vary across different enterprises. This section explores how the influence of AI on corporate financial fraud manifests under different conditions of heterogeneity. Specifically, it examines this relationship across three dimensions: 910 management structure differences (i.e., dual0role vs. non-dual-role governance), (2) external audit supervision (i.e., Big Four vs. non-Big Four auditors), and (3) firm size (i.e., large vs. small enterprises).

5.5.1. Heterogeneity test based on corporate organizational structure

The role of AI may differ between firms with a dual-role structure (where the CEO and Chairman are the same person) and those with a non-dual-role structure. In dual-role firms, the concentration of power in a single individual may result in an imbalance of authority, leading to a lack of effective oversight and weakening of internal checks and balances (Krause et al., 2016). Under such conditions, management may be more prone to abuse of power, directly impacting the integrity of financial reporting and the quality of decision-making processes, thereby facilitating fraudulent behavior (Caiffa et al., 2021). Furthermore, although combining the roles of CEO and Chairman can enhance decision-making efficiency and reduce communication costs, it may simultaneously raise decision-making risks due to insufficient internal discussion and lack of critical review. The pressure to make quick decision can increase the likelihood of oversight regarding the accuracy and transparency of financial disclosure, thus creating room for fraud activities. Additionally, the governance structure of dual-role firms can affect power distribution and decision-making dynamics (Mubeen et al., 2020), which in turn may contribute to the incidence of financial fraud and audit failures. In contrast, firms with a non-dual-role structure, where the CEO and chairman positions are held by separate individuals, benefit from more effective checks and balances. The Chairman is better positioned to independently supervise the management and objectively evaluate the CEO's performance and decisions, instead of simultaneously acting as both the executive and evaluator (Afif et al., 2024). This separation can reduce the potential for abuse of power mitigates the risk of financial fraud.

Building on this theoretical foundation, this study investigates the relationship between AI adoption and financial fraud under different management structures. Specially, the sample is categorized based on whether the roles of CEO and Chairman are merged (*Dual = 1*) or not (*Dual = 0*), forming a dual-role group and a non-dual-role group. Respectively. Separate regressions are conducted for both fraud severity (*Fdeg*) and fraud occurrence (*Fdum*). The regression results presented in Table 7. The findings indicate that AI has a significant and negative impact on financial fraud in non-dual-role firms, suggesting a strong preventive effect. In contrast, the impact of AI is statistically insignificant in firms with a dual-role structure. Specifically, within non-dual-role organizations, AI adoption is more effective in reducing both the likelihood and severity of financial fraud. (See Table 8.)

Therefore, in organizations with a non-dual-role structure, it is crucial for companies to place greater emphasis on the integration of AI technologies in financial oversight processes. This includes the implementation of efficient and automated systems for data analysis, real-time monitoring, and risk detection, which can reduce subjectivity in human decision-making and improve the overall effectiveness of internal supervision, ultimately decreasing the opportunity for financial fraud. However, in dual-role organizations, where the same individual controls both operational and oversight functions, the deterrent effect of AI may be significantly compromised. In such environments, AI systems, if not properly governed, could potentially be misused as tools to conceal fraud, such as through data manipulation, biased algorithmic modeling, or information leakage. These risks highlight the need for strong AI governance frameworks to prevent the exploitation of intelligent systems by powerful individuals within the organization.

5.5.2. Heterogeneity test based on the external regulatory environment of enterprises

From the perspective of an enterprise's external regulatory environment, differences in external audit services may also affect the

Table 7
Results of Heterogeneity Test Based on Organizational Structure.

variables	(1)	(2)	(3)	(4)
	Non-Dual-Role Structure	Dual-Role Structure	Non-Dual-Role Structure	Dual-Role Structure
	Fdeg		*Fdum*	
AI	−0.022***	0.005	−0.075***	−0.027
	(−4.67)	(−0.69)	(−4.58)	(−1.06)
Control	yes	yes	yes	yes
Industry	yes	yes	yes	yes
Year	yes	yes	yes	yes
N	18,111	7332	18,111	7332

Table 8
Results of Heterogeneity Test Based on External Supervision.

variables	(1)	(2)	(3)	(4)
	Non-Big Four	Big Four	Non-Big Four	Big Four
	Fdeg		*Fdum*	
AI	−0.015***	0.004	−0.064***	−0.049
	(−3.69)	(−0.3)	(−4.57)	(−0.62)
Control	yes	yes	yes	yes
Industry	yes	yes	yes	yes
Year	yes	yes	yes	yes
N	23,814	1626	23,815	1628

likelihood of financial fraud. Firms audited by the Big Four accounting firms are generally considered less likely to commit financial fraud. Firstly, the Big Four possess abundant resources and extensive professional expertise. This allows them to allocate more time and deploy highly skilled auditing techniques and data analytics methods, which enhance their ability to detect misstatements and assess fraud risks more effectively. Foer example, PwC's GL.ai robotic solution leverages AI and machine learning techniques to rapidly process large volumes of financial data, identifying potential anomalies and fraud indicators. This technology has been successfully implemented across numerous countries and audit projects (Vitali and Giuliani, 2024). Secondly, as globally recognized providers of audit services, the Big Four are subject to constant scrutiny by regulators, investors, and the public. This external pressure compels them to maintain high standards of professional ethics, audit independence, and regulatory compliance during the audit process (Westermann et al., 2019; Gold et al., 2022). However, even the Big Four are not immune to audit failures. A notable example is PwC's failure in the Evergrande Real Estate, which highlights the limitations of audit assurance, even among top-tier firms.

To further verify whether difference in external audit quality affect the relationship between AI adoption and financial fraud, this study conducts a grouped regression analysis. Firms are divided into two groups based on whether they are audited by one of the four major accounting firms (*Big4 = 1*) or not (*Big4 = 0)*. Separate regressions are conducted for both for the fraud degree (*Fdeg*) and fraud behavior (*Fdum*) with the results presented in Table 7. The regression results indicate that AI significantly inhibits financial fraud in enterprises not audited by the Big Four, suggesting that in the absence of strong external audit mechanisms, internal AI applications play a more vital role in preventing fraudulent behavior. In contrast, the effect in enterprises audited by the Big Four, the coefficients are not statistically significant, indicating a weaker or insignificant incremental effect of AI in these contexts.

Therefore, non-Big Four audit firms should seek to enhance their audit quality to ensure objectivity, independence, and credibility. Concurrently, enterprises themselves must take proactive steps to leverage AI technologies to strengthen internal audits and risk-control systems. Firms should actively explore the use of AI, big data, and other related digital technologies to achieve full-process control, automatic early warning, and traceable oversight in financial operations. This transformation, from "human based control" to "technology-enabled control," can help ensure the timeliness, independence, and effectiveness of internal auditing, thereby improving fraud detection and prevention in financial reporting.

5.5.3. Heterogeneity test based on the nature of the Enterprise itself

Due to the "high-risk, high-investment" nature of AI technologies (Taherdoost and Drazenovic, 2017), firms of different sizes ten to exhibit distinct characteristics in governance structure, resource allocation, and technology application strategies. These differences may influence the extent to which AI can effectively curb financial fraud. Large companies typically possess more sophisticated governance frameworks and richer resources endowments, enabling them to more comprehensively use of AI technologies to enhance internal control and risk management. Compared to smaller companies, large companies are more likely to integrate AI across multiple function areas. For instance, they may apply AI to optimize data processing and supervision procedures. Through large-scale data analysis and automated decision-support system, AI can effectively identify anomalies and potential financial fraud behaviors. Moreover, large companies usually have greater financial capacity and talent reserves, which enhance the depth and breadth of AI application in fraud prevention. Conversely, smaller companies often face constraints in both capital and human resources, limiting their ability to adopt and maintain complex AI systems. This restricts their capacity to apply AI effectively in fraud-risk identification

Table 9
Results of the Heterogeneity Test on Company Size.

variables	(1)	(2)	(3)	(4)
	Companies with small scale	Companies with large scale	Companies with small scale	Companies with large scale
	Fdeg		*Fdum*	
AI	−0.009*	−0.021***	−0.001	−0.012***
	(1.27)	(−3.43)	(−0.45)	(−3.21)
Control	yes	yes	yes	yes
Industry	yes	yes	yes	yes
Year	yes	yes	yes	yes
N	12,883	12,620	12,883	12,620

and internal oversight. In addition, smaller firms may have weaker governance structures and internal control mechanisms, further reducing the efficacy of AI in these contexts.

To test the validate of above conjectures, this paper conducts grouped regressions using the company's size variable. Specifically, companies are divided into large-sized (top 50 %) and small-sized (bottom 50 %) groups based on the distribution of firm size. Separate regressions are performed for both fraud degree (*Fdeg*) and fraud occurrence (*Fdum*), with the results reported in Table 9. The results indicate that AI has a more significant and robust effect in reducing financial fraud in large-scale companies, while in small-scale companies. These findings empirically support the conjecture regarding firm-size heterogeneity.

The regression results in Table 9 indicate that AI can more effectively curb financial fraud in large-scale companies, whereas its effect is relatively weaker in small-scale companies. Considering the typical decision-making and management patterns in companies of different scales, small companies may rely more heavily on the subjective judgment and experience-based decisions of key executives, rather than on structured, data-driven decision-support tools. In such an environment, even if AI technologies are introduced, they may be used more for routine operational support rather than for strategic oversight or risk management.

Therefore, for small-scale companies, additional efforts should be made to enhance both technology application and resource investment. These firms should focus on integrating AI into internal control systems and improving the penetration and efficiency of AI applications. Furthermore, they should explore AI solutions that are tailored to their specific business scale and operational needs to strengthen fraud prevention capabilities. At the same time, large-scale companies should further optimize their AI application strategies. They should aim to ensure that AI technologies are comprehensively cover key aspects of risk identification, real-time monitoring, and governance, so as to maximize their effectiveness in fraud deterrence.

6. Conclusion and policy recommendations

In recent years, as AI technology has developed at a breakneck pace, its application in enterprises has attracted increasing attention. Industries are actively exploring the role of AI across diverse fields, and its impact on daily business operations and broader societal systems cannot be ignored. This study investigates whether, how, and under what conditions corporate AI adoption influences the occurrence of financial fraud. Based on panel data from A-share listed companies in China spanning from 2013 to 2023, this study systematically examines the relationship between AI usage and corporate financial fraud, along with the potential underlying mechanisms. The key findings are as follow: Firstly, the degree of AI adoption significantly reduces the incidence of financial fraud within enterprises. This inhibitory effect is particularly evident in enterprises with separate CEO and Chairman roles in those audited by non-Big Four accounting firms. Second, the mediating analysis reveals that AI primarily affects financial fraud mainly through two critical mechanisms: enhancement of information transparency and strengthening of internal control systems. AI enables firms to quickly process and analyze large-scale data, promoting timely and accurate financial disclosure. It also supports the automation of internal monitoring, risk assessment, and decision-support processes, contributing to more effective fraud detection and prevention. Third, This mitigating effect is especially pronounced in firms with separated chairman and CEO roles, those audited by non-Big Four firms, and larger enterprises.

The research of this paper holds significant implications for both corporate management and external policymakers. In order to fully leverage the benefits of AI in financial fraud prevention and promote its innovative application, the following recommendations are proposed: more effectively utilize the role of AI in curbing corporate financial fraud and driving innovative applications of AI across diverse fields, based on the research findings, the following suggestions are put forward:

Based on the enterprise level:

(1) Increase investments in AI systems, particularly within finance and audit departments. Firms should actively enhance the automation and intelligence degree of financial data processing to reduce human-induced errors and fraudulent opportunities.
(2) Strengthen information transparency by ensuring that financial operations and stakeholder-relevant data are disclosed in a clear, timely. And accessible manner. This is particularly important for enterprises with combined CEO-Chairman roles, where reisks of information manipulation are higher. Enhancing disclosure quality improves stakeholder trust and strengthens both internal and external oversight.
(3) Establishing and maintain robust internal control systems, leveraging AI for risk monitoring, anomaly detection, and compliance assessment. In firms not audited by the Big Four, strong internal controls can compensate for weaker external supervision and reduce fraud risks.

Based on the policymaker level:

(1) Foster a supportive external environment for AI development in enterprises by introducing preferential policies, such as tax incentives, financial subsidies, and low-interest loans, to lower the threshold for AI technology adoption, especially in small and medium-sized enterprises.
(2) As AI-based financial regulation is still in its exploratory phase, policymakers should introduce encourage cross-sector collaboration among government agencies, regulatory bodies, and enterprises. Joint efforts should be made to integrate AI technologies into corporate finance, risk management, and regulatory frameworks.
(3) Refine disclosure standards and strengthen the mandatory information disclosure system. Policymakers should encourage enterprises to apply tools to improve the accuracy and efficiency of their reporting. At the same time, stricter penalties should be

imposed for deliberate concealment, misguidance, or provision of false information to increase the risk and cost for fraud and promote a more transparent and trustworthy business environment.

CRediT authorship contribution statement

Guanglin Sun: Writing – original draft, Methodology, Formal analysis, Conceptualization. **Zhencheng Ling:** Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation. **Yanru Li:** Writing – review & editing. **Chang Xie:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Data curation.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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